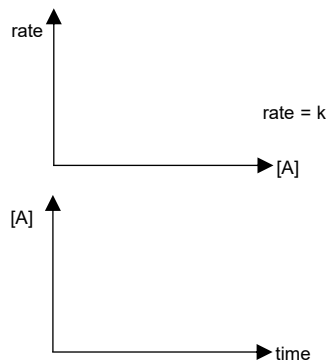


Graphical representations

Consider the following overall reaction: $A \rightarrow 2B$

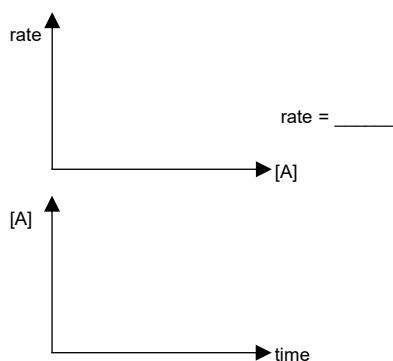
zero order w.r.t. [A]

rate is independent of [A]



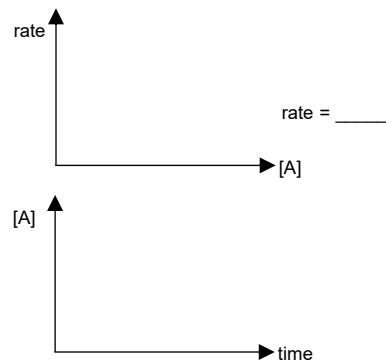
1st order w.r.t. [A]

rate $\propto$ _____



2nd order w.r.t. [A]

rate $\propto$ _____



Rate equation

A reaction has the following overall equation:
 $A + B \rightarrow 2C$

The reaction was found to be 1st order w.r.t. A and 2nd order w.r.t. B.

➤ Write the rate equation for the reaction.

➤ Hence, deduce the units for the rate constant.

➤ What is the overall order of the reaction?

Note:

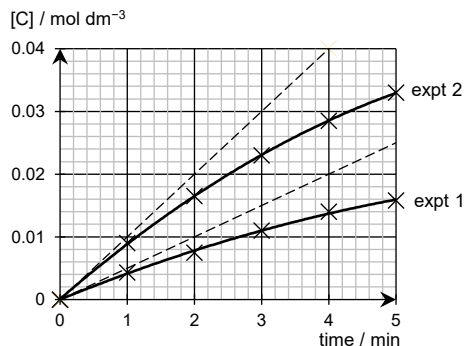
➤ From the overall eqⁿ,
rate of reaction = $-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \left(\frac{1}{2}\right)\frac{d[C]}{dt}$

Common methods to deduce order of reaction

#1 compare initial rate

A reaction has the following overall equation:
 $A + B \rightarrow C$

expt	[A] initial / mol dm ⁻³	[B] initial / mol dm ⁻³	initial rate / mol dm ⁻³ min ⁻¹
1	0.05	0.05	
2	0.10	0.05	
3	0.10	0.10	0.040



Deduce the order of reaction w.r.t. A and B.

❖ initial rate of expt 1:

❖ initial rate of expt 2:

❖ Comparing expt 1 & 2,
When _____
Hence, rate $\propto$ [A]¹, reaction is _____ order w.r.t. A

❖ Comparing expt 2 & 3,
When _____
Since reaction is first order w.r.t A,
Hence, rate $\propto$ [B]², reaction is _____ order w.r.t. B

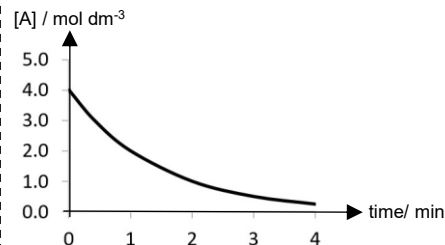
Note:

- draw tangent at $t = 0$ to find initial rate
- rate is always a _____ value

#2 check for constant $t_{1/2}$

A reaction has the following overall equation:
 $A + B \rightarrow C$

B was used in excess.
Deduce the order of reaction w.r.t. A



1st $t_{1/2} \approx$ 2nd $t_{1/2} =$ _____ min

Since $t_{1/2}$ is _____,

reaction is _____ order w.r.t. A

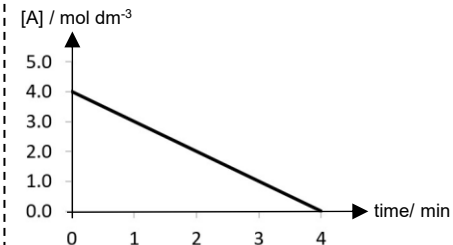
Note:

- draw and label all construction lines clearly
- constant $t_{1/2}$ can also be deduced from a product-time graph

#3 shape of graph

A reaction has the following overall equation:
 $A + B \rightarrow C$

B was used in excess.
Deduce the order of reaction w.r.t. A



Since [A] decreases at a constant rate,
rate of reaction is _____

Reaction is _____ order w.r.t. A

More on $t_{1/2}$

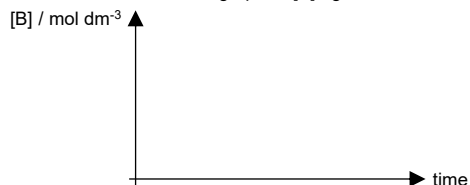
Consider the following reaction: $A \rightarrow 2B$

The reaction is 1st order w.r.t. A.

➤ Complete the table below.

time	[A]/ mol dm ⁻³	[B]/ mol dm ⁻³
initial	0.80	0.00
after 1 st $t_{1/2}$		
after 2 nd $t_{1/2}$		
after 3 rd $t_{1/2}$		

➤ Sketch and label the graph of [B] against time



Relation between k , k' and $t_{1/2}$

Note:

➤ $t_{1/2}$ expression can only be obtained for such rate eqⁿ:

$$\text{rate} = k[A]$$

OR

$$\text{rate} = k'[A]$$

A reaction has the following rate equation

$$\text{rate} = k[A][B][C]$$

➤ Given that A is a catalyst, and $[B] \gg [C]$, express the rate equation in terms of k'

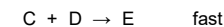
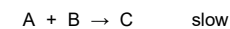
➤ Write an expression for the $t_{1/2}$ of the reaction

➤ Hence state how $t_{1/2}$ changes when [B] doubles

Deducing rate eqⁿ from mechanism

Consider the following reaction: $A + B + D \rightarrow E$

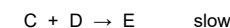
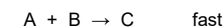
The mechanism involves the following steps



Based on the information, what is the rate equation?

Consider the following reaction: $A + B + D \rightarrow E$

The mechanism involves the following steps



Based on the information, what is the rate equation?

Boltzmann distribution

Temperature

With the aid of a Boltzmann distribution curve, explain why does increasing temperature lead to an increase in rate of reaction.



- ❖ as T increases from T_1 to T_2 , average K.E. of reactant particles increases
- ❖
- ❖ frequency of effective collision increases
- ❖ rate of reaction increases, rate constant increases

Catalyst

With the aid of a Boltzmann distribution curve, explain why does using a catalyst lead to an increase in rate of reaction.



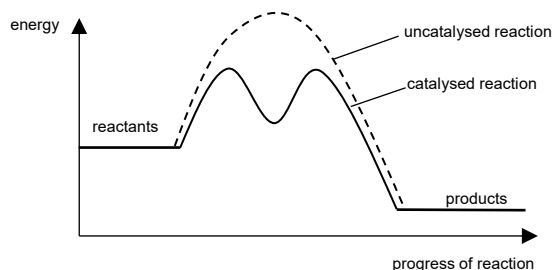
- ❖ catalyst provides an alternative pathway for the reaction with lower E_a
- ❖ more particles have $E \geq E_a$ (catalysed)
- ❖
- ❖ rate of reaction increases, rate constant increases

Catalysis

Homogenous catalyst

the reactants and catalyst are in _____ phase

- Label the transition states and intermediate of the catalysed reaction



essential examples

- Write equations to show how Fe^{2+} catalyses the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$

Step 1: $2\text{Fe}^{2+}(\text{aq}) + \text{S}_2\text{O}_8^{2-}(\text{aq}) \longrightarrow 2\text{Fe}^{3+}(\text{aq}) + 2\text{SO}_4^{2-}(\text{aq})$

Step 2: $2\text{Fe}^{3+}(\text{aq}) + 2\text{I}^-(\text{aq}) \longrightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{I}_2(\text{aq})$

Overall: $\text{S}_2\text{O}_8^{2-}(\text{aq}) + 2\text{I}^-(\text{aq}) \longrightarrow 2\text{SO}_4^{2-}(\text{aq}) + \text{I}_2(\text{aq})$
- Write equations to show how NO_2 catalyses the oxidation of SO_2

Step 1: $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \longrightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$

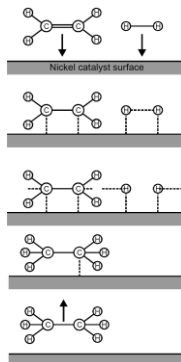
Step 2: $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{NO}_2(\text{g})$

Overall: $\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{SO}_3(\text{g})$

Heterogenous catalyst

the reactants and catalyst are in _____ phase

- Describe the mode of action of a heterogenous catalyst



- ❖ Reactant molecules _____ to catalyst surface and are _____ to active sites of the catalyst
- ❖ The adsorption weakens the covalent bonds within the reactant molecules which lowers activation energy, increases concentration of reactant molecules at catalyst surface and allows reactant molecules to come in close contact with proper orientation.
- ❖ Product molecules _____ and _____ away from the catalyst surface.

essential examples

- Write an equation to describe the reaction in Haber process

$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$$
- Write equations to describe reaction which how the catalytic converter remove oxides of nitrogen, CO and unburnt hydrocarbons C_xH_y .

$$2\text{NO}(\text{g}) + 2\text{CO}(\text{g}) \xrightarrow{\text{Rh}} 2\text{CO}_2(\text{g}) + \text{N}_2(\text{g})$$

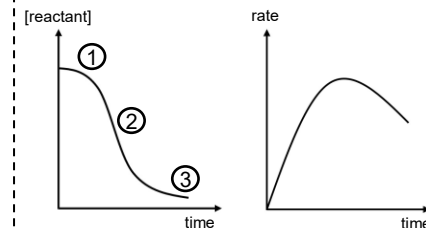
$$\text{NO}_2(\text{g}) + 2\text{CO}(\text{g}) \xrightarrow{\text{Rh}} 2\text{CO}_2(\text{g}) + \frac{1}{2}\text{N}_2(\text{g})$$

$$2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{\text{Pt/Pd}} 2\text{CO}_2(\text{g})$$

$$\text{C}_x\text{H}_y(\text{g}) + \left(x + \frac{y}{4}\right) \text{O}_2(\text{g}) \xrightarrow{\text{Pt/Pd}} x\text{CO}_2(\text{g}) + \frac{y}{2} \text{H}_2\text{O}(\text{g})$$

Unique catalysts

Autocatalyst:



- ❖ a reaction where the product is also the catalyst
- ❖ reaction is _____ at position (1) because _____
- ❖ reaction is _____ at position (2) because _____
- ❖ reaction is _____ at position (3) because _____

Enzymes:

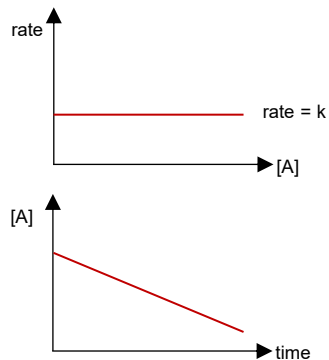
- ❖ Enzymes are _____ molecules which act as _____ catalyst
- ❖ Enzymes have highly _____, where it catalyses only one particular type of reaction. This is due to the complementary shape between the _____ and the _____ of the enzyme.
- ❖ Enzymes are sensitive to conditions: _____ and _____

Graphical representations

Consider the following overall reaction: $A \rightarrow 2B$

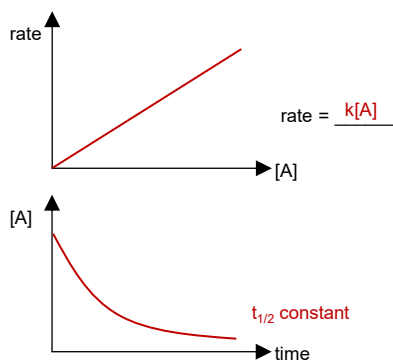
zero order w.r.t. [A]

rate is independent of [A]



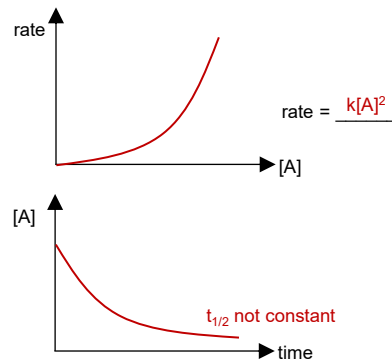
1st order w.r.t. [A]

rate $\propto$ [A]



2nd order w.r.t. [A]

rate $\propto$ [A]²



Rate equation

A reaction has the following overall equation:
 $A + B \rightarrow 2C$

The reaction was found to be 1st order w.r.t. A and 2nd order w.r.t. B.

➤ Write the rate equation for the reaction.

$$\text{rate} = k[A][B]^2$$

➤ Hence, deduce the units for the rate constant.

$$\text{units: mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$$

$$\text{OR mol}^{-2} \text{ dm}^6 \text{ min}^{-1}$$

➤ What is the overall order of the reaction?

3

Note:

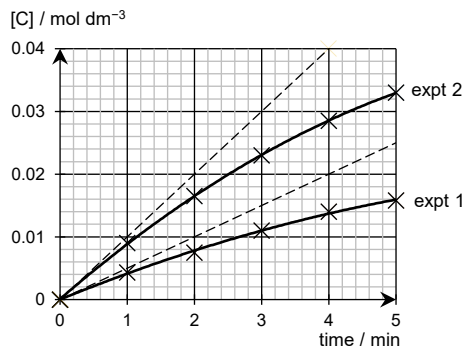
➤ From the overall eqⁿ,
rate of reaction = $-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \left(\frac{1}{2}\right)\frac{d[C]}{dt}$

Common methods to deduce order of reaction

#1 compare initial rate

A reaction has the following overall equation:
 $A + B \rightarrow C$

expt	[A] initial / mol dm ⁻³	[B] initial / mol dm ⁻³	initial rate / mol dm ⁻³ min ⁻¹
1	0.05	0.05	0.005
2	0.10	0.05	0.010
3	0.10	0.10	0.040



Deduce the order of reaction w.r.t. A and B.

- initial rate of expt 1:
gradient of tangent = $\frac{0.02 \text{ mol dm}^{-3}}{4 \text{ min}}$
 $\Rightarrow \text{rate} = 0.005 \text{ mol dm}^{-3} \text{ min}^{-1}$
- initial rate of expt 2:
gradient of tangent = $\frac{0.04 \text{ mol dm}^{-3}}{4 \text{ min}}$
 $\Rightarrow \text{rate} = 0.010 \text{ mol dm}^{-3} \text{ min}^{-1}$

- Comparing expt 1 & 2,
When [A] x 2, rate x 2
Hence, rate $\propto$ [A]¹, reaction is 1st order w.r.t. A (spell out)
- Comparing expt 2 & 3,
When [B] x 2, rate x 4
Since reaction is first order w.r.t. A,
Hence, rate $\propto$ [B]², reaction is 2nd order w.r.t. B (spell out)

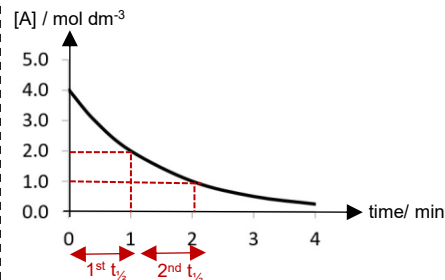
Note:

- draw tangent at $t = 0$ to find initial rate
- rate is always a positive value

#2 check for constant $t_{1/2}$

A reaction has the following overall equation:
 $A + B \rightarrow C$

B was used in excess.
Deduce the order of reaction w.r.t. A



$1^{\text{st}} t_{1/2} \approx 2^{\text{nd}} t_{1/2} = 1 \text{ min}$
Since $t_{1/2}$ is approximately constant,
reaction is 1st order w.r.t. A

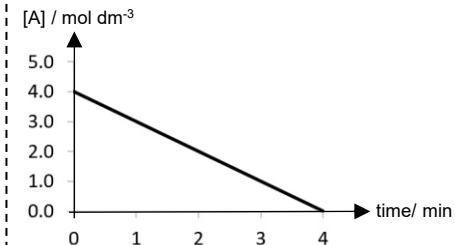
Note:

- draw and label all construction lines clearly
- constant $t_{1/2}$ can also be deduced from a product-time graph

#3 shape of graph

A reaction has the following overall equation:
 $A + B \rightarrow C$

B was used in excess.
Deduce the order of reaction w.r.t. A



Since [A] decreases at a constant rate,
rate of reaction is independent of [A]
Reaction is zero order w.r.t. A

More on $t_{1/2}$

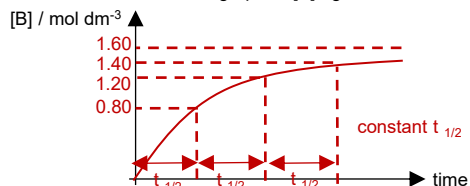
Consider the following reaction: $A \rightarrow 2B$

The reaction is 1st order w.r.t. A.

➤ Complete the table below.

time	[A]/ mol dm ⁻³	[B]/ mol dm ⁻³
initial	0.80	0.00
after 1 st $t_{1/2}$	0.40	0.80
after 2 nd $t_{1/2}$	0.20	1.20
after 3 rd $t_{1/2}$	0.10	1.40

➤ Sketch and label the graph of [B] against time



Relation between k , k' and $t_{1/2}$

Note:

➤ $t_{1/2}$ expression can only be obtained for such rate eqⁿ:

$$\text{rate} = k[A] \quad \text{OR} \quad \text{rate} = k'[A]$$

A reaction has the following rate equation
rate = $k[A][B][C]$

➤ Given that A is a catalyst, and $[B] \gg [C]$, express the rate equation in terms of k'

$$\text{rate} = k'[C]; \quad k' = k[A][B]$$

During the reaction, [A] and [B] will remain approx. constant.

➤ Write an expression for the $t_{1/2}$ of the reaction

$$t_{1/2} = \frac{\ln 2}{k'} \\ = \frac{\ln 2}{k[A][B]}$$

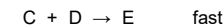
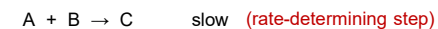
➤ Hence state how $t_{1/2}$ changes when [B] doubles

$t_{1/2}$ halves

Deducing rate eqⁿ from mechanism

Consider the following reaction: $A + B + D \rightarrow E$

The mechanism involves the following steps



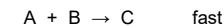
Based on the information, what is the rate equation?

$$\text{rate} \propto [A][B]$$

$$\text{rate} = k[A][B]$$

Consider the following reaction: $A + B + D \rightarrow E$

The mechanism involves the following steps



Based on the information, what is the rate equation?

$$\text{rate} \propto [C][D]$$

$$\propto [A][B][D]$$

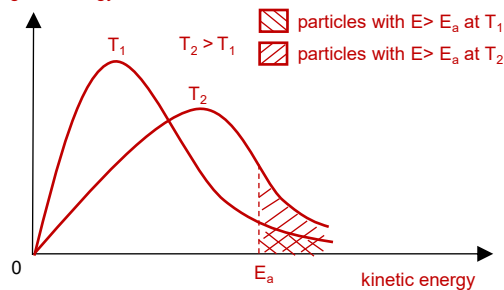
$$\text{rate} = k[A][B][D]$$

Boltzmann distribution

Temperature

With the aid of a Boltzmann distribution curve, explain why does increasing temperature lead to an increase in rate of reaction.

no. of particle w
given energy

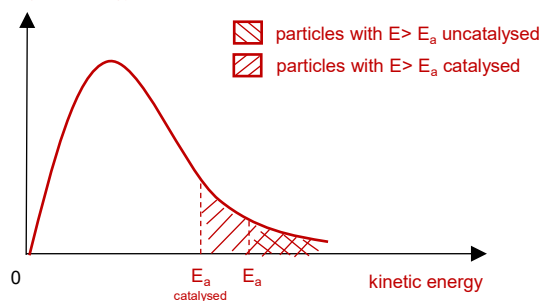


- ❖ as T increases from T_1 to T_2 , average K.E. of reactant particles increases
- ❖ more particles have $E \geq E_a$
- ❖ frequency of effective collision increases
- ❖ rate of reaction increases, rate constant increases

Catalyst

With the aid of a Boltzmann distribution curve, explain why does using a catalyst lead to an increase in rate of reaction.

no. of particle w
given energy



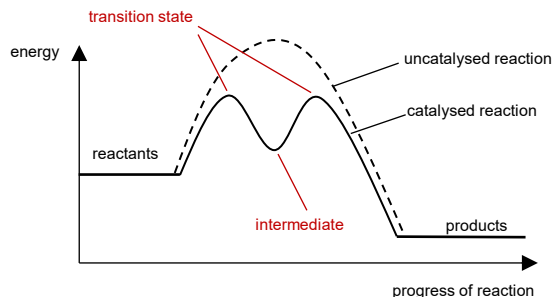
- ❖ catalyst provides an alternative pathway for the reaction with lower E_a
- ❖ more particles have $E \geq E_a$ (catalysed)
- ❖ frequency of effective collision increases
- ❖ rate of reaction increases, rate constant increases

Catalysis

Homogenous catalyst

the reactants and catalyst are in same phase

- Label the transition states and intermediate of the catalysed reaction



essential examples

- Write equations to show how Fe^{2+} catalyses the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$

Step 1: $2\text{Fe}^{2+}(\text{aq}) + \text{S}_2\text{O}_8^{2-}(\text{aq}) \longrightarrow 2\text{Fe}^{3+}(\text{aq}) + 2\text{SO}_4^{2-}(\text{aq})$

Step 2: $2\text{Fe}^{3+}(\text{aq}) + 2\text{I}^-(\text{aq}) \longrightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{I}_2(\text{aq})$

Overall: $\text{S}_2\text{O}_8^{2-}(\text{aq}) + 2\text{I}^-(\text{aq}) \longrightarrow 2\text{SO}_4^{2-}(\text{aq}) + \text{I}_2(\text{aq})$
- Write equations to show how NO_2 catalyses the oxidation of SO_2

Step 1: $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \longrightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$

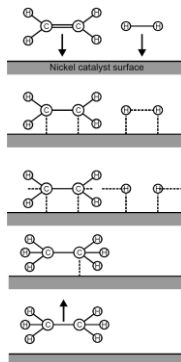
Step 2: $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{NO}_2(\text{g})$

Overall: $\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{SO}_3(\text{g})$

Heterogenous catalyst

the reactants and catalyst are in different phase

- Describe the mode of action of a heterogenous catalyst



- ❖ Reactant molecules diffuse to catalyst surface and are adsorb to active sites of the catalyst
- ❖ The adsorption weakens the covalent bonds within the reactant molecules which lowers activation energy, increases concentration of reactant molecules at catalyst surface and allows reactant molecules to come in close contact with proper orientation.
- ❖ Product molecules desorb and diffuse away from the catalyst surface.

essential examples

- Write an equation to describe the reaction in Haber process

$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$$
- Write equations to describe reaction which how the catalytic converter remove oxides of nitrogen, CO and unburnt hydrocarbons C_xH_y .

$$2\text{NO}(\text{g}) + 2\text{CO}(\text{g}) \xrightarrow{\text{Rh}} 2\text{CO}_2(\text{g}) + \text{N}_2(\text{g})$$

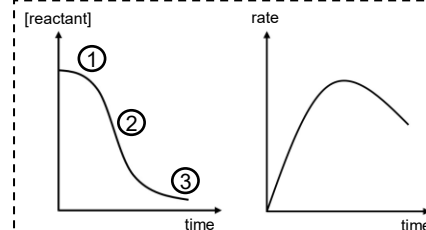
$$\text{NO}_2(\text{g}) + 2\text{CO}(\text{g}) \xrightarrow{\text{Rh}} 2\text{CO}_2(\text{g}) + \frac{1}{2}\text{N}_2(\text{g})$$

$$2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{\text{Pt/Pd}} 2\text{CO}_2(\text{g})$$

$$\text{C}_x\text{H}_y(\text{g}) + \left(x + \frac{y}{4}\right) \text{O}_2(\text{g}) \xrightarrow{\text{Pt/Pd}} x\text{CO}_2(\text{g}) + \frac{y}{2} \text{H}_2\text{O}(\text{g})$$

Unique catalysts

Autocatalyst:



- ❖ a reaction where the product is also the catalyst
- ❖ reaction is slow at position (1) because low [product], therefore low [catalyst]
- ❖ reaction is fast at position (2) because higher [catalyst]
- ❖ reaction is slow at position (3) because low [reactant]

Enzymes:

- ❖ Enzymes are protein molecules which act as biological catalyst
- ❖ Enzymes have highly specific, where it catalyses only one particular type of reaction. This is due to the complementary shape between the substrate and the active site of the enzyme.
- ❖ Enzymes are sensitive to conditions: temperature and pH